Final

1129

Code : 20EC52I

Register Number | | | | | | | | | | |

V Semester Diploma Examination, June/July-2023

INDUSTRIAL INTERNET OF THINGS

Time : 3 Hours] **[Max. Marks : 100**

Instruction : Answer **one** full question from each section.

SECTION – I

1. (a) Dell technologies realized significant cost savings from the use of a predictive maintenance application run in the DELL IT data Lake. The company avoided more than $ 100 million in costs due to this change. Explain the technology of predictive maintenance used in detail. **10**

   (b) In the digital transformation of the energy sector, smart grids and smart meters will be key drivers for a successful digitalization. Explain the necessity of adopting IOT technology in the energy sector. **10**

2. (a) In health care sector, IOT has brought significant changes to improve the facilities and information system during COVID-19 Pandemic. Justify your answer, how IOT has helped in digitization of medical processes and proper management in the hospitals. **10**

   (b) Manufacturing enterprises are seeking to leverage IT/OT convergence to achieve gains in productivity and reduction in cost, while managing the inherent risks of convergence. Discuss the challenges of IT/OT convergence in the Manufacturing Industry. **10**

SECTION – II

3. (a) Which is the most preferred protocol for Car Sensors ? It should support event driven message transfer and queuing through wireless networks. Explain its features and working principle. **10**

(b) Bluetooth offers device control through a smart phone. But, it only allows eight devices to be connected, Zigbee supports upto 65000 nodes, but there is a fee for every connected device. Compare and explain both the technologies. **10**

4. (a) This protocol was not initially created for IOT applications, except for banking environments. In this protocol delivery is guaranteed as well as acknowledgement. Explain the features of this Advanced message queuing protocol. **10**

(b) Amazon side walk is a IOT network that uses LoRa radios to connect devices with Amazon's cloud when they are beyond the reach of traditional home networks. Elaborate the features and concept of the LoRa WAN technology used by Amazon side walk. **10**

SECTION – III

5. (a) The organizations today have to constantly evaluate their cloud infrastructure to ensure that the cloud delivers results, because there are different security and management demands for each of the three cloud models. Explore the features of all the three cloud models in detail. **10**

(b) An IT company decides to provide free access to a public cloud dedication to higher education. Explain all the three cloud computing services it can embrace. **10**

6. (a) The best way to ensure card data security is by using a payment gateway while handling card payments. Explain how gateways are used for data management, local applications and device management in IOT. **10**

(b) In Samsung mobile phones, for setting up the apps stacked in its store, the company is using AWS services, using AWS app development services, Samsung wanted to provide its customer with the facility to download the apps anywhere without any network traffic. List the applications and features of services provided by AWS. **10**

SECTION – IV

7. (a) IOT devices make up for 30% of all network-connected end points, making many companies the primary targets for cyber criminals : Is security support on IOT devices essential ? State a better way to secure the IOT devices. **10**

(b) A security auditing wants to check the security level of IOT based industries. What methodology is incorporated to check the degree of security level ? What are the advantages for the IOT industries by doing so ? **10**

8. (a) IOT based data acquisition system is a collection of software and hardware that allows one to measure or control physical characteristics of something in the real world. Analyse how physical data is measured and controlled in real world using data acquisition system with a neat block diagram. **10**

(b) In the health care industry (Medical IOT) devices are already being utilized to remotely monitor patient's vital signs and has proven very helpful since the pandemic, what are the possible consequences of IOT security attacks in medical IOT ? **10**

SECTION – V

9. (a) In smart asset tracking, system sensors like RFID sensors, GPS, accelerometer sensors etc. monitors required system parameters. The data is recorded in the system and historical data can be viewed whenever required using cloud. In order to design this system, what are the components required and if the system behaves improperly, how do you test and troubleshoot to make the system functional ? **10**

(b) Despite the pandemic induced reduction in air travel, the IATA forecast predicts 8.2 billion Air travellers in 2037. To handle this exponential growth in the number of the passengers in the coming years, Airlines will require the powerful capabilities of Airport Digitalization. Discuss the rise of IOT solutions and their impact on airport experiences. **10**

10. (a) To establish the connectivity between building A computers in one LAN network with building B computers in another LAN through a router. Though after following all the proper procedure like assigning IP addresses to those computers, IP address to Router, proper cable connectivity etc, still the packet of information is not transferring between two LAN. Find the problem and troubleshoot it. **10**

(b) Novartis is one of the world's largest health care companies that provide solutions for patient's well-being. Data analysts of Novartis are taking advantage of the AWS services and still implementing new solutions for the patients. Recommend the data analysis and AWS technology adopted by Novartis to meet its desired requirements and improve the efficiency. **10**

V Semester Diploma Examinations,June/July 2023
Scheme of Valuation

Sub Name – Industrial Internet of Things
Sub Code – 20EC52I
NOTE:

1.Answer any one full question from each section.
2.Each question carries 10 marks
3.All the answers are taken from public domain. If the valuer finds any relevant answer or any nearest relevant points, marks can be awarded.

.....

Section I

1(a).PdM Definition + PdM Principle + PdM Advantages =4 + 3 +3 = 10

1(b).Smart grid Definition + Smart meter Definition + Necessity of IoT = 3 + 3 + 4 = 10

2(a). Any relevant 05 points = 2 x 5 = 10M

2(b).IT/OT Definition + Any relevant 04 Challenges = 2 + 2 X 4 = 02 + 08 = 10 M

Section II

3(a). Identifying MQTT + Principle of MQTT + Any relevant 04 features =2+4+4=10M

3(b). Any relevant 05 comparisons = 2 X 5 = 10M

4(a). Any relevant 05 features = 2 X 5 = 10M

4(b). Any relevant 05 features = 2 X 5 = 10M

Section III

5(a). List 03 Models + Any relevant features of all the 03 Models = 01 + 3 X 3 = 10

5(b). List 03 cloud services + Any relevant features of all the 03 cloud services = 01 + 3 X 3 = 10

6(a).Gateway definition + Any 03 relevant uses in Data management + Any 03 relevant Local applications + Any 03 relevant uses in Device management = 01 +03+03+03=10

6(b). Any relevant 05 features = 2 X 5 = 10

Section IV

7(a). Any relevant 05 security methods = 2 X 5 = 10

7(b). Any relevant 05 methodologies = 2 X 5 = 10

8(a). Block diagram + Explanation of 03 components = 04 + 02 X 03 = 10

8(b). Any relevant 05 consequenses = 2 X 5 = 10

Section V

9(a). Explain any 02 relevant components + Any 03 relevant testing methods = 2 X 2 + 3 X 2 = 10

9(b). Any relevant 05 consequenses = 2 X 5 = 10

10(a). Gateway Explanation + Any 03 relevant troubleshooting methods = 04 + 3 x 2 = 10

10(b). Any relevant 05 technologies = 2 X 5 = 10

V Semester Diploma Examinations,June/July 2023
Model Answer Paper

Sub Name – Industrial Internet of Things
Sub Code – 20EC52I

NOTE:

1.Answer any one full question from each section.
2.All the answers are taken from public domain. If the valuer finds any relevant answer or any nearest relevant points, marks can be awarded.

.....

SECTION-1

1.(a)Dell technologies realized significant cost savings from the use of a predictive maintenance application run in the DELL IT data Lake. The company avoided more than $100 million in costs due to this change. Explain the technology of predictive maintenance used in detail. (PdM Definition + PdM Principle + PdM Advantages = 4 + 3 +3 = 10)

ANS: Predictive maintenance is an application run in the Dell IT data lake, an agile storage platform that can be easily configured for any given data model, structure, application or query.

Predictive maintenance (PdM) is a technique that uses data analysis tools and techniques to detect problems in the operation and possible defects in equipment and processes so you can fix them before they result in failure.

This application leveraged data from sensors on connected devices, lab test results, and vendor technical specifications to determine the likelihood of a system drive failure based on factors ranging from usage to data center environmental conditions. Dell used this information to proactively replace drives that were predicted to fail in the near future. The company avoided more than $100 million in costs associated with repeatedly bringing technicians into its data centers for one-off fixes on failed drives.

To remain competitive and achieve higher margins, exploring how to implement predictive maintenance is likely the right call for most businesses. This refers to the maintenance in which companies leverage their past performance data and all the information that is generated by IoT or sensors. Leveraging this type of predictive analysis can help avoid costly unnecessary repairs, while optimizing the utilization of the equipment and drastically reducing downtime. Dell has identified the value and has started to shift from preventive to predictive maintenance.

Advantages of Predictive Maintenance:

1.IoT based predictive maintenance uses statistical process control principles to determine the best time to perform asset or machine maintenance across industries.

2.It also helps to increase safety and reduce the number of accidents in a factory plant by early detection of faulty equipment or design flaw in a machinery.

3.Predictive maintenance presents you with the best time to work on an asset so that maintenance frequency is minimal and reliability is as high as possible while eliminating unnecessary costs.

4.Predictive maintenance can help companies save money by reducing downtime and preventing equipment failures, it helps companies improve their customer service by allowing them to plan for equipment failures.

.....

(b)In the digital transformation of the energy sector, smart grids and smart meters will be key drivers for a successful digitalization. Explain the necessity of adopting IOT technology in the energy sector. (Smart grid Definition + Smart meter Definition + Necessity of IoT = 3 + 3 +4 = 10)

ANS : The adoption of Internet of Things (IoT) technology in the energy sector is necessary for the successful digitalization of the industry, as it enables the deployment of smart grids and smart meters, which are key drivers of this transformation.

Smart grids are modern, digital electricity grids that use advanced sensors, data analytics, and machine learning algorithms to optimize the flow of electricity and improve the reliability, security, and efficiency of the grid.

Smart meters, on the other hand, are advanced metering devices that can measure and report on electricity usage in real-time, providing utilities with detailed data on customer energy consumption patterns.Necessity of using IoT technology:

1.Smart grids and smart meters can be connected and controlled remotely,

2.IoT will enable utilities to gather real-time data on grid performance, identify problems, and take corrective action.

3.IoT can help to improve the reliability and efficiency of the grid, reduce energy waste, and lower operating costs.

4.The deployment of smart grids and smart meters can also benefit consumers, as it enables them to monitor their own energy usage, identify ways to reduce their energy consumption, and potentially save money on their energy bills.

5.The adoption of IoT technology in the energy sector is essential for the successful digitalization of the industry, as it enables the deployment of advanced, data-driven systems that can improve the performance, reliability, and efficiency of the energy grid.

.....

2.(a) In health care sector, IOT has brought significant changes to improve the facilities and information system during COVID-19 Pandemic. Justify your answer, how IOT has helped in digitization of medical processes and proper management in the hospitals. (Any relevant 05 points = 2 x 5 = 10M)

ANS : The Internet of Things (IoT) has played a significant role in the digitization of medical processes and proper management in hospitals, particularly during the COVID-19 pandemic. Some examples of how IoT has helped in this regard include:

1.Telemedicine: loT devices, such as wearable devices and smart devices, have enabled the remote monitoring of patients, allowing doctors to remotely diagnose and treat patients from a distance.

2.This has been particularly important during the pandemic, as it has allowed hospitals to reduce the number of in-person visits and minimize the risk of infection transmission.

3.Remote patient monitoring: IoT devices, such as smart watches and smart sensors, have also enabled the remote monitoring of patient's vital signs, such as heart rate, blood pressure, and oxygen levels. This has allowed hospitals to monitor patients remotely and take prompt action if their condition deteriorates.

4.Inventory management: IoT devices, such as RFID tags and sensors, have been used to track the movement of medical equipment and supplies within hospitals. This has helped hospitals to better manage their inventory, ensuring that the right equipment is available in the right place atthe right time.

5.Asset tracking: IoT devices, such as GPS trackers, have been used to track the movement of hospital assets, such as ambulances and medical equipment. This has helped hospitals to better manage their resources and respond more quickly to patient needs.

Overall, the adoption of IoT technology in the healthcare sector has brought significant changes to improve facilities and information systems, and has played a vital role in the digitization of medical processes and proper management in hospitals.

.....

(b) Manufacturing enterprises are seeking to leverage IT/OT convergence to achieve gains in productivity and reduction in cost, while managing the inherent risks of convergence. Discuss the challenges of IT/OT convergence in the Manufacturing Industry.

(IT/OT Definition + Any relevant 04 Challenges = 2 + 2 X 4 = 02 + 08 = 10 M)

ANS: The convergence of Information Technology (IT) and Operational Technology (OT) in the manufacturing industry refers to the integration of IT systems, which handle data and communications, with OT systems,which control and monitor physical processes.

This convergence can enable manufacturers to achieve gains in productivity and cost reduction

by enabling the integration and analysis of data from both IT and OT systems. However, there are several challenges that manufacturers must consider when pursuing IT/OT convergence:

1.Security risks: The integration of IT and OT systems can create new security vulnerabilities, as it exposes OT systems to the potential risks associated with IT systems, such as cyber attacks and data breaches. Manufacturers must therefore invest in robust security measures to protect their systems and data.

2.Compatibility issues: IT and OT systems may use different technologies, protocols, and standards, which can make it difficult to integrate them. Manufacturers must ensure that their systems are compatible and can communicate with each other effectively.

3.Integration complexity: Integrating IT and OT systems can be complex and time-consuming, as it requires the integration of different systems, data sources, and processes. Manufacturers must ensure that they have the resources and expertise to effectively manage the integration process.

4.Change management: Converging IT and OT systems can require significant changes to processes and procedures, which can be difficult for manufacturers to manage. It is important that manufacturers have a clear plan in place to manage the transition to IT/OT convergence and ensure that all stakeholders are on board.

Overall, IT/OT convergence in the manufacturing industry can bring significant benefits, but it also presents a range of challenges that manufacturers must carefully consider and address in order to realize these benefits.

.....

SECTION-2

3.(a) Which is the most preferred protocol for Car Sensor? It should support event driven message transfer and queuing through wireless networks. Explain its features and working principle.(Identifying MQTT + Principle of MQTT + Any relevant **04 features =2+4+4=10M)**

ANS: One of the most preferred protocols for IoT devices is MQTT .It collects data from various electronic devices and supports remote device monitoring. It is a subscribe/publish protocol that runs over Transmission Control Protocol (TCP), which means it supports event-driven message exchange through wireless networks. MQTT is mainly used in devices which are economical and requires less power and memory. For instance, fire detectors, car sensors, smart watches, and apps for text-based messaging.The salient features of MQTT protocol are as follows:

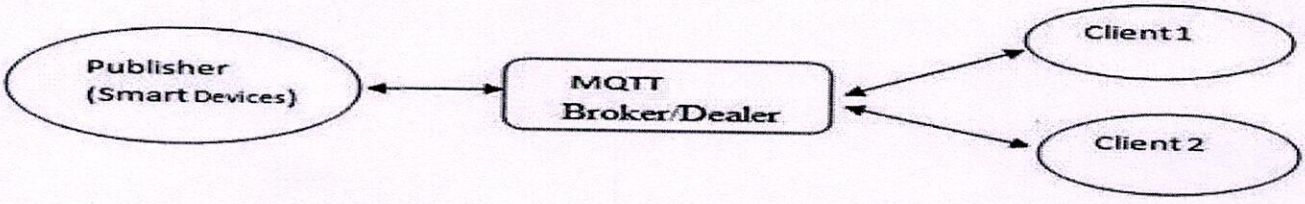
1.Machine to machine communication can be done with this MQTT. It was developed by IBM.

2.Message queue telemetry transport protocol is a messaging protocol. This protocol collects the data from the devices and forwards to the network. So the connection between the devices and network can be done by this protocol.

3.It's a simple protocol which sends the data from the sensors to devices and then towards the network.

4.This is the top protocol of the TCP/IP protocol reference model. The three elements in MQTT protocol in IoT are subscriber, publisher and dealer/broker.

5.The data can be interchanged between subscriber and publisher. The dealer/broker enables the safety connection between the subscriber and publisher.



.....

(b) Bluetooth offers device control through a Smartphone. But, it only allows eight devices to be connected, Zigbee supports upto 65000 nodes, but there is a fee for every connected device. Compare and explain both the technologies. (Any relevant 05 comparisons = 2 X 5 = 10M)

ANS: Bluetooth and ZigBee are both wireless communication technologies that are commonly used for device control and automation applications. However, they have some key differences in terms of their capabilities and characteristics:

1.Range: Bluetooth has a shorter range than ZigBee, typically up to 30 meters (100 feet) in open space. ZigBee, on the other hand, has a longer range, up to 100 meters (300 feet) in open space.

2.Device support: Bluetooth allows up to eight devices to be connected to a single master device, while ZigBee supports up to 65,000 nodes (devices) in a single network.

3.Data rate: Bluetooth has a higher data rate than ZigBee, typically up to 2.1 Mbps. ZigBee has a lower data rate,typically up to 250 Kbps.

4.Power consumption: Bluetooth requires more power than ZigBee, as it has a higher data rate and a shorter range. This can make it less suitable for battery-powered devices. ZigBee, on the other hand, is designed for low-power consumption, making it suitable for use in devices that require long battery life.

5.Cost: Bluetooth is typically free to use, as it is an open standard. ZigBee, on the other hand, charges a fee for every connected device, as it is a proprietary standard.

Overall, Bluetooth and ZigBee are both useful technologies for device control and automation applications, but they have different capabilities and characteristics that make them more suitable for certain applications.

.....

4.(a) This protocol was not initially created for IOT applications, except for banking environments. In this protocol delivery is guaranteed as well as acknowledgement. Explain the features of this Advanced message queuing protocol.

(Any relevant 05 features = 2 X 5 = 10M)

ANS: The Advanced Message Queuing Protocol (AMQP) is an emerging standard for message-oriented middleware (MOM). MOM supports high quality-of-service, asynchronous program-to-program communication between two or more components of an application, or between two or more applications. The salient features of AMQP are as follows:

1.Guaranteed delivery: AMQP ensures that messages are delivered reliably, even if the recipient is offline or unavailable. If a recipient is unable to receive a message, the message is stored in a queue until it can be delivered.

2.Acknowledgment: AMQP allows recipients to acknowledge the receipt of a message, which helps to ensure that the sender knows that the message has been received.

3.Routing: AMQP allows messages to be routed to specific recipients based on predefined rules, which can help to improve the efficiency of message delivery.

4.Security: AMQP supports the use of encryption and authentication to secure messages and ensure that they are only delivered to the intended recipients.

5.Inter-operability: AMQP is designed to be interoperable with other messaging protocols, allowing it to be used in a variety of different systems and environments.

Overall, AMQP is a feature-rich messaging protocol that is well-suited for use in loT applications, due to its ability to ensure the reliable delivery and acknowledgment of messages.

.....

(b) Amazon sidewalk is a IOT network that uses LoRa radios to connect devices with Amazon's cloud when they are beyond the reach of traditional home networks. Elaborate the features and concept of the LoRaWAN technology used by Amazon sidewalk.

(Any relevant 05 features = 2 X 5 = 10M)

ANS: LoRaWAN is a long-range, low-power wireless networking technology that is used by Amazon Sidewalk to connect Internet of Things (IoT) devices to Amazon's cloud when they are beyond the reach of traditional home networks. Some key features and concepts of LoRaWAN include:

1.Range: LoRaWAN has a long range, typically up to 15 km (9.3 miles) in urban environments and up to 40 km (25 miles) in rural environments. This makes it suitable for use in loT applications where devices may be located at a distance from the network infrastructure.

2.Low power consumption: LoRaWAN is designed for low-power consumption, making it suitable for use in devices that require long battery life.

3.Security: LoRaWAN uses advanced encryption and authentication techniques to secure communications and prevent unauthorized access to the network.

4.Frequency bands: LoRaWAN operates in unlicensed frequency bands, which means that it can be used without the need for a license. However, the specific frequency bands that it uses vary by region.

5.Network architecture: LoRaWAN uses a star-of-stars network architecture, in which devices communicate with "gateways" that are connected to the network infrastructure. This allows LoRaWAN to support a large number of devices while minimizing the infrastructure required.

Overall, LoRaWAN is a powerful and flexible technology that is well-suited for use in loT applications, particularly in cases where devices may be located at a distance from the network infrastructure and need to operate for long periods of time on battery power.

.....

SECTION-3

5.(a) The organizations today have to constantly evaluate their cloud infrastructure to ensure that the cloud delivers results, because there are different security and management demands for each of the three cloud models. Explore the features of all the three cloud models in detail.
(List 03 Models +Any relevant Features of all the 03 Models = 01 + 3 X 3 = 10)

ANS: There are three main models for cloud computing:

1.Public Cloud
2. Private Cloud
3.Hybrid Cloud.

Each of these models has its own unique features and characteristics, and organizations must carefully evaluate their needs and requirements in order to determine which model is the most suitable for their needs.

Public cloud:

A public cloud is a cloud computing model in which resources, such as servers, storage, and applications, are provided by a third-party provider and made available to the public over the Internet. Public clouds are typically owned and operated by large companies, such as Amazon Web Services (AWS) and Microsoft Azure. Some key features of public clouds include:

1.Pay-as-you-go pricing: Public clouds typically charge users on a pay-per-use basis, allowing organizations to only pay for the resources they consume. Scalability: Public clouds are highly scalable, allowing organizations to easily increase or decrease their resource usage as needed.

2.No upfront costs: Public clouds do not require organizations to make upfront investments in hardware or software, as all resources are provided by the provider.

Private cloud:

A private cloud is a cloud computing model in which resources are dedicated to a single organization and are not shared with other organizations. Private clouds can be owned and operated by the organization themselves, or they can be provided by a third-party provider.Some key features of private clouds include:

1.Customization: Private clouds can be customized to meet the specific needs of an organization.

2.Security: Private clouds offer higher levels of security, as resources are not shared with other organizations.

3.Control: Private clouds give organizations more control over their resources, as they are not subject to the policies and procedures of a third-party provider.

Hybrid cloud:

A hybrid cloud is a cloud computing model that combines elements of both public and private clouds, allowing organizations to use a combination of on-premises and off-premises resources. Some key features of hybrid clouds include:

1.Flexibility: Hybrid clouds allow organizations to choose the resources and services that are best suited to their needs, whether they are provided by a public cloud provider or hosted on-premises.

2.Cost savings: Hybrid clouds can help organizations to reduce costs by allowing them to take advantage of the pay-as-you-go pricing model of public clouds while still maintaining control over sensitive data and applications through the use of private clouds.

3.Scalability: Hybrid clouds allow organizations to scale their resources up or down as needed, depending on their changing needs.

Overall, each of the three cloud models has its own unique features and characteristics, and organizations must carefully evaluate their needs and requirements in order to determine which model is the most suitable for their needs.

.....

(b) An IT company decides to provide free access to a public cloud dedication to higher education. Explain all the three cloud computing services it can embrace.

(List 03 cloud services + Any relevant features of all the 03 cloud services = 01 + 3 X 3 = 10)

ANS: There are three main types of cloud computing services:

1.Infrastructure as a Service (IaaS),

2. Platform as a Service (PaaS),

3. Software as a Service (SaaS).

An IT company that decides to provide free access to a public cloud dedicated to higher education could embrace any or all of these services, depending on the needs and requirements of the institutions that will be using the cloud.

Infrastructure as a Service (IaaS):

IaaS is a cloud computing model in which the cloud provider offers access to a range of infrastructure resources, such as servers, storage, and networking, over the Internet. Higher education institutions could use laaS to access the computing resources they need for research, teaching, and other activities, without having to invest in their own infrastructure.

Platform as a Service (PaaS):

PaaS is a cloud computing model in which the cloud provider offers access to a platform for developing, testing, and deploying applications. Higher education institutions could use PaaS to develop and deploy custom applications for use in research, teaching, and other activities.

Software as a Service (SaaS):

SaaS is a cloud computing model in which the cloud provider offers access to software applications over the Internet. Higher education institutions could use SaaS to access a range of applications, such as productivity tools, learning management systems, and research tools, without having to install and maintain the software on their own systems.

Overall, by offering access to IaaS, PaaS, and/or SaaS services, an IT company could provide higher education institutions with a range of cloud computing resources that they can use to support their research, teaching, and other activities.

.....

6.(a) The best way to ensure card data security is by using a payment gateway while handling card payments. Explain how gateways are used for data management, local applications and device management in IOT.

(Gateway definition + Any 03 relevant uses in Data management +Local applications + Device management = 01 +03+03+03=10)

ANS:A payment gateway is a service that enables businesses to securely accept and process card payments from customers. In the context of the Internet of Things (IoT), payment gateways can be used for data management, local applications, and device management in several ways:

Data management:

Payment gateways can help to ensure the security and integrity of card data by encrypting and storing it securely. This can help to prevent unauthorized access to sensitive data and protect against data breaches.

Local applications:

Payment gateways can be integrated with local applications, such as point-of-sale systems, to

enable card payments to be processed directly from the application. This can help to streamline the payment process and reduce the risk of errors.

Device management:

Payment gateways can be used to manage and secure devices that are connected to the IoT, such as card readers and point-of-sale terminals. This can help to ensure that only authorized devices are able to access the payment gateway and process card payments.

Overall, payment gateways can play a key role in ensuring the security and integrity of card data in IoT applications, as well as supporting the management of local applications and devices.

.....

(b) In Samsung mobile phones, for setting up the apps stacked in its store, the company is using AWS services, using AWS app development services, Samsung wanted to provide its customer with the facility to download the apps anywhere without any network traffic. List the applications and features of services provided by AWS.

(Any relevant 05 features = 2 X 5 = 10)

ANS: Amazon Web Services (AWS) provides a wide range of cloud computing services and tools to help businesses and organizations build, deploy, and manage applications at scale. Some of the services and features provided by AWS include:

1.Compute: Services like Amazon EC2 (Elastic Compute Cloud) and AWS Lambda allow you to run and scale applications in the cloud.

2.Storage: Services like Amazon S3 (Simple Storage Service) and Amazon EBS (Elastic Block Store) provide scalable and secure storage options for your data. Databases: Services like Amazon RDS (Relational Database Service) and Amazon DynamoDB provide managed database solutions for your applications.

3.Networking: Services like Amazon VPC (Virtual Private Cloud) and Amazon Route 53 allow you to create and manage networks in the cloud.

4.Security: Services like Amazon IAM (Identity and Access Management) and Amazon GuardDuty help you secure your applications and data in the cloud.

5.Management tools: Services like AWS CloudFormation and AWS CodePipeline allow you to automate the process of deploying and managing your applications.

6.Analytics: Services like Amazon Athena and Amazon EMR (Elastic MapReduce) allow you to analyze and process large datasets in the cloud.

7.Machine learning: Services like Amazon SageMaker and Amazon Rekognition allow you to build and deploy machine learning models in the cloud.

8.Internet of Things (IoT): Services like Amazon IoT Core and Amazon FreeRTOS allow you to build and manage IoT applications in the cloud.

.....

SECTION-4

7.(a) IOT devices make up for 30% of all network-connected end points,making many companies the primary targets for cyber criminals: Is security support on IOT devices essential? State a better way to secure the IOT devices. (Any relevant 05 security methods = 2 X 5 = 10)

ANS: Yes, security support on loT devices is essential because they are increasingly being targeted by cybercriminals. IoT devices often have limited computing resources and are not designed with security in mind, making them vulnerable to cyber attacks.The following methods can be used to secure the IoT devices :

1.One way to better secure loT devices is to ensure that they are regularly patched and updated with the latest security fixes.

2.It is also important to use strong, unique passwords for each device and to use encryption to protect the data transmitted between the device and the network.

3.Another important security measure is to implement network segmentation, which involves separating a lot of devices from other parts of the network to limit their access and reduce the potential attack surface.

4.In addition, it is important to use a firewall and intrusion detection/prevention system (IDPS) to protect against network-based attacks and to monitor for unusual activity.

5.Finally, it is important to educate users about the importance of IoT device security and to provide them with guidelines for securely using and managing these devices.

.....

(b) A security auditing wants to check the security level of IOT based industries. What methodology is incorporated to check the degree of security level? What are the advantages for the IOT industries by doing so? (Any relevant 05 methodologies = 2 X 5 = 10)

ANS: Security auditing is an important step in ensuring the security of IoT-based industries. There are several methodologies that are commonly used to check the security levels of IoT structures, including:

1.vulnerabilities in an IoT system: This can include identifying known vulnerabilities in the software and hardware used in the system, as well as any potential weaknesses in the system's design or configuration.

2.Vulnerability assessments: A vulnerability assessment is used to identify and analyze potential security

3.Penetration testing: Penetration testing simulates a real-world attack on an IoT system to identify potential security vulnerabilities. This can include attempting to exploit known vulnerabilities or attempting to access sensitive data.

4.Risk assessments: A risk assessment is used to identify and evaluate potential risks to an IoT system. This can include assessing the potential impact of a security breach on the system and the organization, as well as identifying any potential threats and vulnerabilities.

5.Compliance assessments: IoT industries are required to comply with different regulations, standard and laws. A compliance assessment is used to check whether an IoT system is in compliance with relevant regulations, standards and laws that affects the IoT industry.

Advantages of conducting security audits are:

6.Identifying vulnerabilities: Security audits can help identify vulnerabilities in an IoT system before they can be exploited by an attacker. This can help organizations take steps to protect against security breaches before they occur.

7.Improving compliance: Compliance audits can help ensure that loT systems are in compliance with relevant regulations, standards, and laws. This can reduce the risk of costly fines and penalties, and can improve the organization's reputation.

8.Reducing Risk: Security audits can help identify potential risks and vulnerabilities in an IoT system. By identifying these risks, organizations can take steps to mitigate them, reducing the overall risk to the system and the organization.

9.Improving overall security: By regularly conducting security audits, companies can improve the overall security posture of their IoT systems and better protect against security breaches and other cyber threats. Cost saving: By identifying and addressing security vulnerabilities before they are exploited, security audits can help organizations avoid the high costs of responding to a security breach.

Overall, conducting security audits on loT systems is a critical step in ensuring the security of these systems. By regularly conducting vulnerability assessments, penetration testing, risk assessments, and compliance assessments, organizations can identify and address potential security vulnerabilities and improve the overall security of their IoT systems.

.....

8.(a) IOT based data acquisition system is a collection of software and hardware that allows one to measure or control physical characteristics of something in the real world. Analyze how physical data is measured and controlled in the real world using a data acquisition system with a neat block diagram.(Block diagram + Explanation of 03 components = 04 + 02 X 03 = 10)

ANS:A data acquisition system typically consists of three main components:

1.Sensors:

These are devices that measure a physical characteristic of something in the real world, such as temperature, pressure, or humidity. The IoT sensor is a device that measures a physical characteristic such as temperature or pressure, and is equipped with connectivity and software that allows it to transmit data over the internet. The lo T gateway is a device that acts as a bridge between the sensor and the cloud server. It may be used to filter and process the data from the sensor before forwarding it to the server, and may also be usedto control the sensor or other devices.

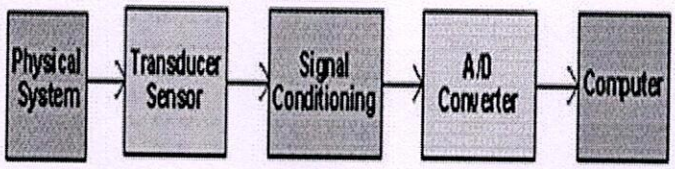
2.Data acquisition hardware:

This is the hardware that connects to the sensors and collects the data from them. It may also include components such as amplifiers or filters to enhance the accuracy of the measurements.

3.Data acquisition software:

This is the software that controls the data acquisition hardware and processes the data collected from the sensors. It may also include visualization tools to display the data in a graphical format. The computer is used to access and analyze the data from the cloud server, and may also be used to control the operation of the data acquisition system. In this example, the lo T sensor and gateway are used to remotely collect and transmit data from the physical system, while the cloud server and computer are used to store, process, and analyze the data.

Here is a block diagram of a typical data acquisition system:



The Internet of Things (IoT) is a network of physical devices, vehicles, buildings, and other objects that are embedded with sensors, software, and connectivity, allowing them to collect and exchange data. In a data acquisition system, IoT devices can be used to remotely monitor and control physical systems, and to collect and transmit data from remote locations.For example, an IoT-enabled temperature sensor could be used to monitor the temperature in aremote warehouse, and transmit the data to a central server where it could be analyzed and used to adjust the heating and cooling systems in the warehouse. In

this way, loT technology can be used to improve the efficiency and effectiveness of a data acquisition system by allowing it to operate remotely and in real-time.

.....

(b) In the healthcare industry (Medical IOT) devices are already being utilized to remotely monitor patient's vital signs and has proven very helpful since the pandemic, what are the possible consequences of IOT security attacks in medical IOT? (Any relevant 05 consequenses = 2 X 5 = 10)

ANS: The consequences of loT security attacks in the medical industry can be severe and even life-threatening. For example, if an attacker were to gain access to a patient's medical device and manipulate the data it is sending. This could lead to incorrect treatment decisions being made by the healthcare providers. Additionally, if an attacker were to gain access to a device that is used to administer medication, they could potentially increase or decrease the dosage, which could have serious consequences for the patient. There is also the risk of sensitive patient data being accessed or stolen in a security breach, which could lead to a violation of patient privacy and trust. The possible consequences of IOT security attacks in medical IoT are as follows

1.Incorrect treatment decisions: If an attacker is able to manipulate the data being sent by a medical device, this could lead to incorrect treatment decisions being made by healthcare providers.

2.Medication errors: If an attacker gains access to a device that administers medication, they could potentially increase or decrease the dosage, which could have serious consequences for the patient.

3.Patient harm: In general, a security breach or attack on a medical loT device could potentially lead to harm or injury for the patient.

4.Patient data breaches: There is also the risk of sensitive patient data being accessed or stolen in a security breach, which could lead to a violation of patient privacy and trust.

5.Loss of trust in the healthcare system: If patients do not feel that their medical devices and data are secure, they may lose trust in the healthcare system as a whole. This could lead to a decrease in the adoption of medical IoT devices and a reluctance to share personal health information.

.....

9.(a) In smart asset tracking, system sensors like RFID sensors, GPS, accelerometer sensors etc. monitors required system parameters. The data is recorded in the system and historical data can be viewed whenever required using the cloud. In order to design this system, what are the components required and if the system behaves improperly, how do you test and troubleshoot to make the system functional? (Explain any 02 relevant components + Any 03 relevant testing methods = 2 X 2 + 3 X 2 = 10)

ANS: Smart asset tracking systems use sensors and other technologies to track the location and status of assets, such as vehicles, equipment, and inventory. Some common types of sensors used in smart asset tracking systems include:

1.GPS sensors: These sensors can be used to determine the location of an asset in real-time, using satellite signals.

2.RFID sensors: These sensors can be used to identify and track assets using radio frequency technology.

3.Accelerometers: These sensors can be used to measure the acceleration of an asset, which can be used to detect movement and changes in velocity.

Testing and troubleshooting Methods :

1.These sensors can be used to monitor the temperature of an asset, which can be useful for detecting problems or ensuring that temperature-sensitive goods are being stored or transported properly.

2.To test these sensors, it is important to ensure that they are functioning correctly and producing accurate readings. This may involve calibrating the sensors, conducting tests with known values, and comparing the sensor readings to expected results.

3.If a sensor is not functioning properly, the first step in troubleshooting it is to identify the issue. This may involve checking for physical damage to the sensor, testing the sensor's connectivity to other systems, and examining the sensor's data output for any anomalies.

4.Once the issue has been identified, appropriate steps can be taken to repair or replace the sensor as needed.

.....

(b) Despite the pandemic induced reduction in air travel, the IATA forecast predicts 8.2 billion Air travelers in 2037. To handle this exponential growth in the number of the passengers in the coming years, Airlines will require the powerful capabilities of Airport Digitalization. Discuss the rise of IOT solutions and their impact on airport experiences.

(Any relevant 05 consequenses = 2 X 5 = 10)

ANS: IoT solutions can play a significant role in the digitization of airports and the management of the growing number of passengers. Some potential applications of IoT in the airport industry include:

1.Asset tracking:

IoT sensors can be used to track the location and condition of assets such as luggage carts, cleaning equipment, and ground support vehicles. This can help to improve the efficiency of airport operations by reducing the time spent searching for these assets and ensuring that they are properly maintained.

2.Queue management:

IoT sensors can be used to track the flow of passengers through security checkpoints, ticketing counters, and other areas of the airport to identify bottlenecks and optimize the allocation of staff and resources.

3.Environmental monitoring:

IoT sensors can be used to monitor environmental conditions such as temperature, humidity, and air quality in the airport to ensure that passengers are comfortable and to identify any potential hazards.

4.Predictive maintenance:

IoT sensors can be used to monitor the condition of equipment such as baggage handling systems, escalators, and elevators to predict when maintenance is needed and prevent downtime.

5.Overall, the use of IoT solutions can help airports to improve efficiency, reduce costs, and enhance the passenger experience.

.....

10.(a) To establish the connectivity between building A computers in one LAN network with building B computers in another LAN through a router. Though after following all the proper procedures like assigning IP addresses to those computers, IP address to Router, proper cable connectivity etc, still the packet of information is not transferring between two LAN. Find the problem and troubleshoot it.

(Gateway Explanation + Any 03 relevant troubleshooting methods = 04 + 3 x 2 = 10)

ANS: The data sent from the source device is not reaching the destination because we have not assigned the router IP address as default gateway address for each end device.A default gateway makes it possible for devices in one network to communicate with devices in another network. In other words, the default gateway is an intermediate device between the local network and the internet.

The default gateway is an important device for the data forwarding and routing of the data on the other network. It helps in the communication of one network computer with the other network computer. So the default gateways of the router should be assigned to all the end devices in the network.in the same way we should set default gateway for other side of the router and assign it to the end devices which is present in that LAN.There could be a number of reasons why the packets of information are not transferring between the two LANS. Here are a few things you can try to troubleshoot the problem:

1.Check the cable connections to make sure they are secure and in good working condition.

2.Make sure that the router is properly configured and has the correct IP addresses for the two LANs. Check to see if there are any firewalls or security protocols that are blocking the transfer of information between the two LANS.

3.Check the status of the network interfaces on the computers to make sure they are active and configured properly. Try pinging the computers in one LAN from the other to see if there is a connection.

4.Check the logs on the router to see if there are any error messages that might indicate the source of the problem.

.....

(b)Novartis is one of the world's largest healthcare companies that provides solutions for patient's well- being. Data Analysts of Novartis are taking advantage of the AWS servicesand still implementing new solutions for the patients. Recommend the data analysis and AWS technology adopted by Novartis to meet its desired requirements and improve the efficiency.

(Any relevant 05 technologies = 2 X 5 = 10)

ANS: As a large healthcare company, Novartis likely has a variety of data analysis and technologyneeds in order to meet the desired requirements of its patients and improve efficiency. Some potential data analysis and AWS technologies that Novartis may be using include:

1.Data lakes and warehouses:

Novartis may be using AWS technologies such as Amazon S3 and Amazon Redshift to create a centralized data lake or data warehouse where it can store and analyze large amounts of structured and unstructured data. This could include data from electronic medical records, clinical trials, research studies, and other sources.

2.Machine learning:

Novartis may be using AWS machine learning services such as AmazonSageMaker to build, train, and deploy machine learning models that can analyze and extract insights from data. These models could be used for a variety of purposes, such as predicting patient outcomes, detecting patterns in large data sets, or identifying trends and anomalies.

3.Business intelligence and visualization:

Novartis may be using AWS tools such as Amazon QuickSight to create interactive dashboards, charts, and reports that help data analysts and decision makers visualize and understand data trends and patterns. These tools can help Novartis gain insights into its operations and make informed decisions.

4.Data processing and ETL:

Novartis may be using AWS technologies such as Amazon Glue and Amazon EMR to extract, transform, and load data from various sources into its data lake or warehouse. These tools can help Novartis clean, enrich, and prepare data for analysis, and automate the data ingestion process.

5.Cloud computing:

Novartis may be leveraging the power and scalability of AWS cloud computing services such as Amazon EC2 and Amazon ECS to run data analysis and machine learning workloads. These services can help Novartis scale its data processing and analysis capabilities as needed, and reduce the cost and complexity of managing on-premises infrastructure.

.....

" I hereby certify that the model answer script prepared by me for the subject code "20EC52IT" is taken from public domain. If the valuer finds any relevant answer or any nearest relevant points, marks can be awarded.

Jyothishree T
Selection Grade Lecturer
Dept of Electronics and Communication
106-D.R.R(Govt)Polytechnic
Davanagere

